INCH-POUND
MIL-DTL-23053/1C
17 June 1996
SUPERSEDING
MIL-I-23053/1B
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, CROSSLINKED CHLORINATED POLYOLEFIN, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053

REQUIREMENTS:

Polymer type: The base elastomer used in formulating this sleeving shall be a crosslinked chlorinated polyolefin.

Continuous operating temperature range:

Class 1: -55°C (-67°F) to +90°C (+194°F)
Class 2: -70°C (-94°F) to +121°C (+250°F)

Color: The sleeving shall be furnished in a black color that conforms to Class II of MIL-STD-104 (see 1.2.1 and 3.4.1.5).

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2a):

Class 1 - Normal operating temperature
Class 2 - Extended operating temperature

NOTICE:

Class 1 sleeving because of its temperature limitations should not be considered for new design applications. Class 2 sleeving can be substituted for Class 1 sleeving.

Longitudinal change: +1, -10 percent

AMSC N/A FSC 5970
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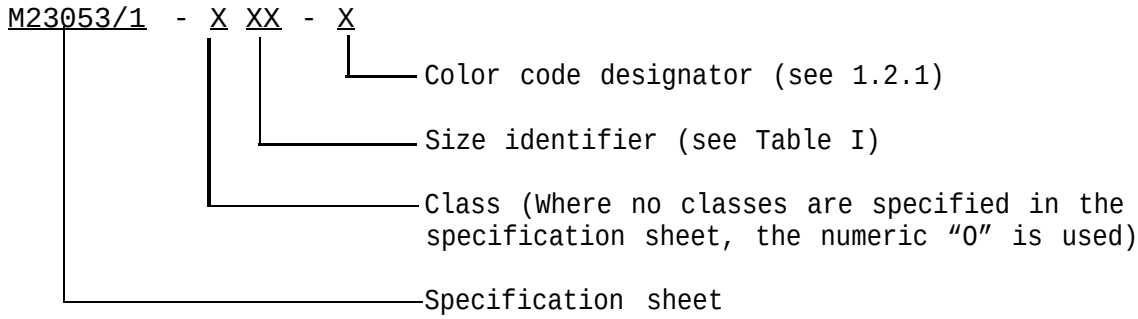
Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers shown as follows:



Example: Class 2, black, 1.000 inch (25.4 mm) as supplied ID sleeving shall be identified as M23053/l-207-0.

TABLE I. Construction details, inches (mm). 1/

Military Part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/1-101-0	.250 (6.4)	.143 (3.6)	.035 ± .010 (.89 ± .25)
M23053/1-102-0	.375 (9.5)	.211 (5.4)	.040 ± .010 (1.01 ± .25)
M23053/1-103-0	.500 (12.7)	.286 (7.3)	.048 ± .015 (1.21 ± .38)
M23053/1-104-0	.625 (15.9)	.357 (9.1)	.052 ± .015 (1.32 ± .38)
M23053/1-105-0	.750 (19.1)	.428 (10.9)	.057 ± .015 (1.44 ± .38)
M23053/1-106-0	.875 (22.2)	.500 (12.7)	.065 ±.020 (1.65 ± .51)
M23053/1-107-0	1.000 (25.4)	.570 (14.5)	.070 ± .020 (1.77 ± .51)
M23053/1-108-0	1.250 (31.8)	.714 (18.1)	.087 ± .020 (2.20 ± .51)
M23053/1-109-0	1.500 (38.1)	.857 (21.8)	.095 ± .020 (2.41 ± .51)
M23053/1-110-0	1.750 (44.5)	1.000 (25.4)	.107 ± .020 (2.71 ± .51)
M23053/1-111-0	2.000 (50.8)	1.140 (29.0)	.110 ± .020 (2.79 ± .51)
M23053/1-112-0	3.000 (76.2)	1.710 (43.4)	.125 ± .020 (3.17 ± .51)
M23053/1-113-0	4.000 (101.6)	2.280 (57.9)	.140 ± .020 (3.55 ± .51)
Class 2			
M23053/1-201-0	.250 (6.4)	.143 (3.6)	.035 ± .010 (.89 ± .25)
M23053/1-202-0	.375 (9.5)	.211 (5.4)	.040 ± .010 (1.01 ± .25)
M23053/1-203-0	.500 (12.7)	.286 (7.3)	.048 ± .015 (1.21 ± .38)
M23053/1-204-0	.625 (15.9)	.357 (9.1)	.052 ± .015 (1.32 ± .38)
M23053/1-205-0	.750 (19.1)	.428 (10.9)	.057 ± .015 (1.44 ± .38)
M23053/1-206-0	.875 (22.2)	.500 (12.7)	.065 ± .020 (1.65 ± .51)

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TABLE I. Construction details, inches (mm). 1/ - Continued

Military Part Number 3/	As supplied ID minimum	After unrestricted Shrinkage	
		ID maximum	Wall thickness 2/
M23053/1-207-0	1.000 (25.4)	.570 (14.5)	.070 ± .020 (1.77 ± .51)
M23053/1-208-0	1.250 (31.8)	.714 (18.1)	.087 ± .020 (2.20 ± .51)
M23053/1-209-0	1.500 (38.1)	.857 (21.8)	.095 ± .020 (2.41 ± .51)
M23053/1-210-0	1.750 (44.5)	1.000 (25.4)	.107 ± .020 (2.71 ± .51)
M23053/1-211-0	2.000 (50.8)	1.140 (29.0)	.110 ± .020 (2.79 ± .51)
M23053/1-212-0	3.000 (76.2)	1.710 (43.4)	.125 ± .020 (3.17 ± .51)
M23053/1-213-0	4.000 (101.6)	2.280 (57.9)	.140 ± .020 (3.55 ± .51)

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The color code identified is the standard acquisition color.

Unrestricted shrinkage procedures: Test method 4.6.5; 175° ± 2°C (347° ± 4°F) for 10 minutes, maximum.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and condition
As supplied:		
ID, minimum	Table I	4.6.3.1.1
Low temperature flexibility	No cracks	4.6.7.1 Class 1:-55° ± 2°C (-67° ± 4°F); Class 2: -70° ± 2°C (-94° ± 4°F)
Heat shock	No cracks, flowing or dripping	4.6.8 Class 1: 150° ± 2°C; (302° ± 4°F); Class 2: 200° ± 2°C (392° ± 4°F)
Restricted shrinkage	No cracks	4.6.6.1.1 135° ± 2°C (275° ± 4°F)
Voltage withstand	Pass	4.6.6.2

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test Procedure and condition
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3.1.2
Wall thickness	Table I	4.6.3.2
Tensile strength, psi (MPa), minimum	1,500 (10.3)	4.6.13 ASTM D412
Ultimate elongation, percent, minimum	225	4.6.13 ASTM D412
Tensile stress, at 200 percent elongation psi (MPa), maximum	1,500 (10.3)	4.6.12.2 ASTM D412
Dielectric strength, volts/roil (Kv/mm), min	300 (11.8) - Up to 0.070 wall thickness 200 (7.8) - 0.070 and above	4.6.2 ASTM D2671
Volume resistivity, Ohm-CM, minimum	1×10^{11}	4.6.2 ASTM D876
Corrosion	No pitting or corrosion	4.6.10.1 and 4.6.10.2 at 150° ± 2°C (302° ± 4°F) for 16 hours.
Water absorption, percent, maximum	1.0	4.6.2 ASTM D570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 Class 1: 100° ± 2°C (212° ± 4°F), 96 hours; Class 2: 121° ± 2°C (250° ± 4°F), 168 hours
Tensile strength, psi (MPa), minimum	1,200	
Ultimate elongation, percent, minimum	Class 1 - 150 Class 2 - 175	

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and condition
Dielectric strength, volt/roil (Kv/mm), minimum	300 (11.8)	4.6.11
Fluid resistance, properties after:	Class 1 - 900 (6.2) Class 2 - 1,000 (6.9)	
Tensile strength, psi (MPa), minimum	Class 1 - 150 Class 2 - 175	
Ultimate elongation, percent, minimum	Class 1 - 200 (7.9) Class 2 - 250 (9.8)	
Dielectric strength, volts/roil (Kv/mm), minimum	Self extinguishing 15 seconds; 3 inches	4.6.14, Procedure A, ASTM D2671
Flammability	No growth	4.6.2 ASTM G21
Fungus resistance		

1/ Unless otherwise specified, requirements, test procedures and conditions are for all classes.

Storage life conditions: Supplier shall certify to storage at 18 to 35°C (65 to 95°F) for 1 year. Conformance to 3.5. for storage life extension.

Intended use: Heat shrinkable flexible polychloroprene sleeving is intended for use on cables or harness systems in equipment that is subject to high levels of physical abuse.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA, MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

(Project 5970-1129-01)

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